

# Statistical Analysis Plan

## Causal Effects of Tissue Microstructure on Diffusion MRI Contrasts in Multiple Sclerosis: A Bayesian Hierarchical Analysis of Postmortem 9.4 T MRI–Histology Data

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### Abstract

This document presents the pre-specified statistical analysis plan for a study linking postmortem diffusion MRI metrics (fractional anisotropy, FA; mean diffusivity, MD) to co-registered histological tissue sections in multiple sclerosis (MS). Thirteen tissue blocks from one MS brain were scanned at 9.4 T with  $200\ \mu\text{m}$  isotropic resolution and stained for five histological markers (myelin, astrocytes, microglia, cell nuclei, axons). The analysis adopts an explicitly causal framework, formalised through directed acyclic graphs (DAGs), to identify the effects of mesoscopic tissue composition on diffusion contrasts. Bayesian hierarchical regression models with tissue-type-specific effects, block-level random intercepts, and heteroscedastic residuals form the modelling backbone. A total of 19 pre-specified models (9 FA regressions + 8 MD regressions + 2 g-computation procedures) test distinct causal estimands. This plan specifies all estimands, identification strategies, likelihoods, priors, sensitivity analyses, and planned outputs prior to the confirmatory analysis of the full 13-block dataset.

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# 1 Introduction and Research Question

## 1.1 Scientific Context

Diffusion MRI is widely used to infer tissue microstructure in the living brain, yet the precise relationship between histological tissue composition and diffusion contrasts remains incompletely characterised. In multiple sclerosis, lesions exhibit heterogeneous combinations of demyelination, axonal loss, gliosis, and inflammation, each of which may independently alter diffusion properties. Understanding these relationships at the mesoscopic scale is essential for interpreting clinical diffusion MRI and for developing microstructure-informed biomarkers.

## 1.2 Study Design

This study uses postmortem tissue from one MS brain, comprising 13 tissue blocks scanned at 9.4 T with 200  $\mu\text{m}$  isotropic resolution. Each block was subsequently sectioned and stained with five histological stains:

- **Myelin** (MBP)
- **Astrocytes** (GFAP)
- **Microglia** (CR343)
- **Cell nuclei** (Hematoxylin)
- **Axons** (Bielschowsky silver impregnation)

Semi-quantitative density maps were derived from each stain at matching resolution. Additionally, a fiber coherence map is available for the FA models, derived from structural tensors of the axon segmentation.

Manually annotated regions of interest (ROIs) classify tissue into 10 distinct types covering normal-appearing white matter, lesion cores, perilesional areas, lesion-surrounding white matter, partially demyelinated white matter, three different grey matter layers, diffuse cortical demyelination and partial cortical demyelination. Data are organized as approximately 20-pixel sub-patches within these ROIs, yielding 4 021 patches across all 13 blocks (based on preliminary data characterization).

## 1.3 Research Question

*What are the causal effects of mesoscopic tissue characteristics (myelin, axons, astrocytes, microglia, cellularity) on diffusion clinically relevant MRI contrasts (MD, FA) in MS tissue, and how do these effects differ across tissue types?*

Key aspects of this formulation:

1. **Both total and direct effects.** Total effects are the primary estimand; direct effects (via mediation decomposition) are secondary where identifiable. For example, the total effect of axon density on FA includes the path mediated through myelin, whereas the direct effect excludes it.
2. **Biophysical scope.** The causal influence of tissue microstructure on diffusion is a physical relationship. With  $N = 1$  brain and 13 tissue blocks spanning diverse tissue types and lesion stages, the study provides intensive biophysical characterisation across different disease contexts within a single donor, rather than population-level prediction.
3. **Regional comparison** Regional analysis is motivated by i) a technical concern that unobserved exposures could vary significantly between tissue regions and is ii) a central question of the study to investigate whether different causal effect drive the diffusion contrasts in different tissue regions.

## 2 Causal Framework: Directed Acyclic Graphs

### 2.1 Rationale for a Causal Approach

Rather than fitting a single regression of FA or MD on all available predictors simultaneously (an approach that conflates direct effects, mediated effects, and collider bias) we construct a directed acyclic graph (DAG) encoding our substantive knowledge of the data-generating process. The DAG serves three purposes:

1. It makes causal assumptions transparent and debatable, allowing collaborators and reviewers to critique the assumed structure rather than an opaque modelling choice.
2. It determines which variables must be conditioned on (and which must not be conditioned on) for each estimand (via the rules of  $d$ -separation).
3. It identifies which causal effects are identifiable from the observed data and which are not, given unmeasured variables.

The DAGs were constructed in **dagitty** and encode biophysical knowledge about how tissue composition influences diffusion MRI signals, as well as the roles of unmeasured confounders and technical factors.

### 2.2 Key Structural Decisions

A single DAG is used for all ROI types; the same biophysical mechanisms operate across tissue compartments, though the magnitudes of individual contributors vary. It is extremely important not to confuse patho-mechanistic causal links from early disease with the biophysical causal links in our observed post-mortem snapshot (e.g. axonal degeneration follows demyelination pathologically, but biophysically the presence of myelin is caused by the existence of axons, not the reverse). The following structural decisions were made:

- **Axon  $\rightarrow$  Myelin, Coherence, Cell, FA:** Axon-Density is the root biological exposure. It directly influences myelin density (myelination depends on axon presence), fibre coherence (axon packing determines local orientation dispersion), total cell density (axons contribute to the cell count and attract oligodendrocytes), and FA directly via the restricted diffusion compartment within axon cylinders.
- **GFAP  $\rightarrow$  Cell and FA:** Astrocyte density (GFAP) is exogenous (no biological parents in the measured snapshot). It contributes cells to the total cell count and directly alters FA via expanded extracellular tortuosity.
- **Microglia  $\rightarrow$  Cell, Iron, and FA:** Microglia-Density is exogenous. It contributes to total cell density, deposits iron (Microglia  $\rightarrow$  Iron  $\rightarrow$  FA, Iron is unmeasured/latent), and directly affects FA via cell-body-restricted diffusion.
- **Oligodendrocytes [latent]  $\rightarrow$  Myelin and Cell:** Oligodendrocyte cell bodies contribute to total cell density and are the primary source of myelin. Oligodendrocytes cannot be separately measured from the available stains (the Hematoxylin cell nuclei count subsumes all cell types), so Oligodendrocytes is a latent common cause of Myelin-Density and Cell-Density. This has critical consequences for causal identification (see Section 2.5).
- **Axon  $\rightarrow$  Coherence  $\rightarrow$  FA:** Fiber coherence mediates part of the axon-density effect on FA. The Myelin  $\rightarrow$  Coherence edge is absent (myelin affects coherence only through its effect on axon integrity, already captured by Axon  $\rightarrow$  Coherence).
- **Cell-Density is a collider** with observed parents {Axon, GFAP, Microglia} and latent parent {Oligodendrocytes}. It must not be included as a covariate alongside its parents when the goal is to estimate those parents' total effects (conditioning on a collider induces spurious associations among its parents). When Cell-Density is the exposure, its parents are confounders and must be controlled for.
- **Iron mediates part of Microglia  $\rightarrow$  FA:** The pathway Microglia  $\rightarrow$  Iron  $\rightarrow$  FA is present but Iron is unmeasured (latent). This prevents identification of the pure direct

effect of Microglia on FA.

- **Hydration-Status [latent]**: Post-mortem tissue hydration is driven by fixation duration and storage time (Fixation Duration  $\rightarrow$  Hydration-Status  $\leftarrow$  Storage Time) and directly affects FA. Including Fixation Duration and Storage Time as technical covariates proxies for this latent variable.
- **ECM [latent]**: Extracellular matrix composition affects FA residually but is not caused by any measured predictor. It adds residual FA variance only.
- **Technical covariates** (Fixation Duration, Storage Time) are constant within a tissue block and are included as fixed-effect covariates in all models; residual block-level variation is absorbed by block-level random intercepts. Post-mortem interval (PMI) is identical for all 13 blocks (24 days) and is therefore excluded (zero variance; unidentifiable).

### 2.3 DAG for FA

Figure 1 shows the DAG for the FA outcome. Measured variables with available histological stains are shown with filled nodes; unmeasured variables are shown with open nodes. Green edges indicate causal paths to FA.

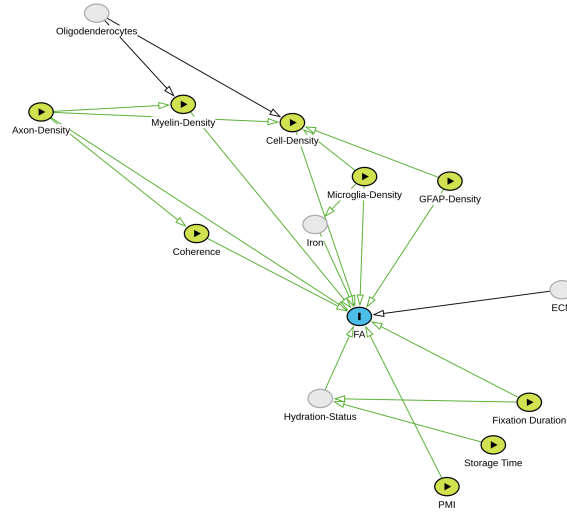


Figure 1: Directed acyclic graph for fractional anisotropy (FA). The outcome node (FA) is shown in blue. Measured biological predictors: Axon-Density, Myelin-Density, Microglia-Density, GFAP-Density, Cell-Density, Coherence. Latent biological variables (unmeasured): Iron, ECM, Hydration-Status, Oligodendrocytes. Technical factors: PMI, Storage Time, Fixation Duration. Crucially, Oligodendrocytes [latent] is a common cause of both Myelin-Density and Cell-Density, creating a confounding structure that must be addressed in the adjustment sets (see Section 2.5).

### 2.4 DAG for MD

The DAG for MD (Figure 2) is identical to the FA DAG with the exception that the Coherence node and its edges are absent, since fiber coherence does not affect mean diffusivity.

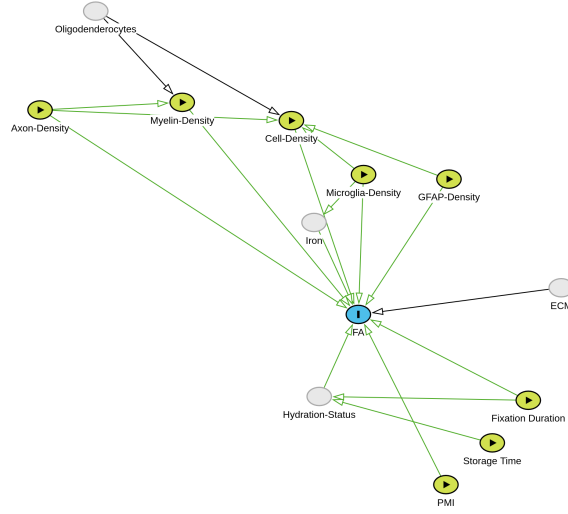


Figure 2: Directed acyclic graph for mean diffusivity (MD). Structure identical to Figure 1 except that the Coherence node is absent.

## 2.5 Identifiable Estimands and Causal Identification

We apply the formal backdoor criterion ( $d$ -separation) to the DAG to derive the minimal adjustment set for each estimand. The latent variable Oligodendrocytes (which causes both Myelin-Density and Cell-Density) is a central identification challenge, as it creates confounding backdoor paths that must be blocked using observed proxy variables. Below we derive adjustment sets for each exposure in turn, then summarize in Tables 1 and 2.

**Notation.** Let **Tech** = {Fixation Duration, Storage Time} be the technical covariates, included in all models for precision (they are not confounders of any biological exposure because they have no causal path from any biological node to FA other than directly; PMI is constant across all blocks and is excluded). Let Oligo denote the latent Oligodendrocytes variable. A path is blocked if it passes through a conditioned non-collider, or if all colliders on the path are unconditioned.

### 2.5.1 Axon-Density $\rightarrow$ FA

Axon-Density has no parents (exogenous), so there are no backdoor paths. The total causal effect is identified by simple regression:

$$F1: \quad FA \sim \text{Axon} + \text{Tech} + \text{RE}.$$

For the controlled direct effect (CDE) of Axon excluding mediators {Myelin, Coherence, Cell}, we condition on all three. Conditioning on Cell (a collider with parents Axon, GFAP, Microglia, Oligo) opens spurious paths and to close them we must additionally condition on GFAP and Microglia (observed) while Myelin (already in **Z**) blocks the Oligo-induced path. The CDE model is:

$$F7: \quad FA \sim \text{Axon} + \text{Myelin} + \text{Coherence} + \text{Cell} + \text{GFAP} + \text{Microglia} + \text{Tech} + \text{RE}.$$

Only  $\hat{\beta}_{\text{Axon}}$  from F7 has a causal interpretation (CDE) and all other coefficients are identification nuisance parameters.

The total indirect effect of Axon (via all three mediators jointly) is:  $\hat{\beta}_{\text{Axon}}^{(\text{F1})} - \hat{\beta}_{\text{Axon}}^{(\text{F7})}$ . Individual mediator contributions require g-computation (F10, see below).

### 2.5.2 Coherence $\rightarrow$ FA

Coherence has one parent (Axon), so all backdoor paths run through Axon. Conditioning on Axon blocks them all:

$$\text{F2: } \text{FA} \sim \text{Coherence} + \text{Axon} + \text{Tech} + \text{RE}.$$

Coherence  $\rightarrow$  FA is the only outgoing edge of Coherence, so there are no mediators and no CDE/NDE decomposition is applicable. Note that this model also yields  $\hat{\beta}_{\text{Axon}}^{(\text{F2})}$ : the Axon effect after partialling out the Coherence-mediated component. The difference  $\hat{\beta}_{\text{Axon}}^{(\text{F1})} - \hat{\beta}_{\text{Axon}}^{(\text{F2})}$  estimates the Axon  $\rightarrow$  Coherence  $\rightarrow$  FA indirect effect (valid under linearity).

### 2.5.3 Myelin-Density $\rightarrow$ FA

Myelin has two parents, Axon (observed) and Oligo (latent). The critical backdoor path is:

$$\text{Myelin} \leftarrow \text{Oligo} \rightarrow \text{Cell} \rightarrow \text{FA}.$$

Cell is a non-collider on this path, so conditioning on Cell blocks it. However, conditioning on Cell (a collider) opens paths among its parents, requiring GFAP and Microglia to be added to  $\mathbf{Z}$ . The full adjustment set is  $\mathbf{Z} = \{\text{Axon}, \text{Cell}, \text{GFAP}, \text{Microglia}\} + \text{Tech}$ , and none of its members are descendants of Myelin (Myelin  $\rightarrow$  FA is the only outgoing edge):

$$\text{F3: } \text{FA} \sim \text{Myelin} + \text{Axon} + \text{Cell} + \text{GFAP} + \text{Microglia} + \text{Tech} + \text{RE}.$$

Only  $\hat{\beta}_{\text{Myelin}}$  from F3 is a causal total effect. The other coefficients are nuisance. Because Myelin has no mediators, total effect = direct effect = CDE.

### 2.5.4 GFAP-Density $\rightarrow$ FA

GFAP has no parents (exogenous). Its total causal effect, with Cell as its sole mediator (GFAP  $\rightarrow$  Cell  $\rightarrow$  FA), is estimated without conditioning on Cell:

$$\text{F4: } \text{FA} \sim \text{GFAP} + \text{Tech} + \text{RE}.$$

The **CDE of GFAP excluding Cell** requires conditioning on Cell and additionally on Axon, Microglia, Myelin (to close collider-induced paths):

$$\text{F8: } \text{FA} \sim \text{GFAP} + \text{Cell} + \text{Axon} + \text{Microglia} + \text{Myelin} + \text{Tech} + \text{RE}.$$

**NDE/NIE for GFAP  $\rightarrow$  Cell  $\rightarrow$  FA is not identified.** Latent Oligo causes both Cell (the mediator) and Myelin  $\rightarrow$  FA (a parallel path not through Cell), violating the no-unmeasured-mediator-outcome-confounder assumption. Any mediation claim for GFAP must be accompanied by sensitivity analysis (Imai residual-correlation parameter  $\rho$ ).

### 2.5.5 Microglia-Density $\rightarrow$ FA

Microglia has no parents (exogenous). Its mediators are Iron (latent) and Cell. Total effect:

$$F5 : \quad FA \sim \text{Microglia} + \text{Tech} + \text{RE}.$$

For the CDE excluding Cell (Iron path remains active), condition on Cell and control for the collider-induced paths via Axon, GFAP, Myelin:

$$F9 : \quad FA \sim \text{Microglia} + \text{Cell} + \text{Axon} + \text{GFAP} + \text{Myelin} + \text{Tech} + \text{RE}.$$

$\hat{\beta}_{\text{Microglia}}$  in F9 captures the Microglia  $\rightarrow$  FA direct path plus Microglia  $\rightarrow$  Iron  $\rightarrow$  FA (since Iron is unmeasured). It is therefore a CDE excluding Cell only, not a pure direct effect. The CDE excluding both Cell and Iron is not identifiable (Iron is latent).

### 2.5.6 Cell-Density $\rightarrow$ FA

Cell has three observed parents (Axon, GFAP, Microglia) and one latent parent (Oligo). The critical backdoor path runs through Oligo:

$$\text{Cell} \leftarrow \text{Oligo} \rightarrow \text{Myelin} \rightarrow \text{FA}.$$

Conditioning on Myelin (a non-collider on this path) blocks it. After conditioning on Myelin (which is a collider between Axon and Oligo), new paths open but are all blocked by conditioning on Axon (already in  $\mathbf{Z}$ ). The adjustment set is  $\mathbf{Z} = \{\text{Axon}, \text{GFAP}, \text{Microglia}, \text{Myelin}\} + \text{Tech}$ :

$$F6 : \quad FA \sim \text{Cell} + \text{Axon} + \text{GFAP} + \text{Microglia} + \text{Myelin} + \text{Tech} + \text{RE}.$$

**Important:** Myelin is included here purely as a confounding-control proxy for latent Oligo, not as a mediator of Cell. Only  $\hat{\beta}_{\text{Cell}}$  is causally interpretable. Cell  $\rightarrow$  FA is the only outgoing edge of Cell, so no mediation decomposition is possible.

### 2.5.7 Axon Mediation: Interventional Effects via G-Computation

For individual-mediator decomposition of the Axon total effect (into contributions via Myelin, Coherence, and Cell), we use parametric Monte Carlo g-computation (interventional direct/indirect effects). This approach is preferred over natural direct/indirect effects (NDE/NIE) because latent Oligo induces correlation between the Myelin and Cell mediators that would violate cross-world independence in the NDE framework.

#### G-computation steps:

1. Fit mediator models (in parallel, since Myelin and Cell have no causal link between them):
  - Myelin  $\sim$  Axon + Tech + RE (Oligo absorbed into residuals)
  - Cell  $\sim$  Axon + GFAP + Microglia + Tech + RE (Oligo absorbed into residuals; adding Myelin as covariate partially accounts for Oligo correlation)
  - Coherence  $\sim$  Axon + Tech + RE
2. Fit outcome model F7 (the full CDE model).
3. Monte Carlo: for each observation  $i$  and draw  $s$ , sample mediator values under Axon =  $x$  and Axon =  $x'$  from the fitted mediator models; predict  $\hat{y}$  under mixed combinations; average to compute interventional direct and indirect effects.
4. Bootstrap the entire procedure (steps 1–3) for credible intervals.

**Key identification caveat:** The Oligo latent variable induces residual correlation between Myelin and Cell that cannot be fully eliminated in the mediator models. Sensitivity analysis using Imai’s  $\rho$  parameter and a “drop-one-mediator” diagnostic are required for all mediation results.



Table 1: Identifiable estimands for FA models. **Tech** = {Fixation Duration, Storage Time} (PMI is constant across all blocks and excluded). All models include a block-level random intercept ( $1 \mid \text{block}$ ) and heteroscedastic residuals  $\sigma \sim \text{ROI}_{\text{type}}$ . ROI-type-specific slopes are estimated via predictor  $\times$  ROI-type interaction terms. The causally interpretable coefficient is shown in the rightmost column; all other coefficients in multi-predictor models are identification nuisance parameters.

| #                                                     | Estimand                                                       | Predictors in model                                  | Adj. set Z                                        | Causal coeff.                                                         |
|-------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|
| F1                                                    | Axon $\rightarrow$ FA (total)                                  | Axon, Tech                                           | Tech only                                         | $\hat{\beta}_{\text{Axon}}$                                           |
| F2                                                    | Coherence $\rightarrow$ FA (total)                             | Coherence, Axon, Tech                                | {Axon} + Tech                                     | $\hat{\beta}_{\text{Coherence}}$                                      |
| F3                                                    | Myelin $\rightarrow$ FA (total)                                | Myelin, Axon, Cell, GFAP, Microglia, Tech            | {Axon, Cell, GFAP, Microglia} + Tech              | $\hat{\beta}_{\text{Myelin}}$ only                                    |
| F4                                                    | GFAP $\rightarrow$ FA (total)                                  | GFAP, Tech                                           | Tech only                                         | $\hat{\beta}_{\text{GFAP}}$                                           |
| F5                                                    | Microglia $\rightarrow$ FA (total)                             | Microglia, Tech                                      | Tech only                                         | $\hat{\beta}_{\text{Microglia}}$                                      |
| F6                                                    | Cell $\rightarrow$ FA (total)                                  | Cell, Axon, GFAP, Microglia, Myelin, Tech            | {Axon, GFAP, Microglia, Myelin} + Tech            | $\hat{\beta}_{\text{Cell}}$ only                                      |
| F7                                                    | Axon $\rightarrow$ FA (CDE, excl. Myelin, Coherence, Cell)     | Axon, Myelin, Coherence, Cell, GFAP, Microglia, Tech | {Myelin, Coherence, Cell, GFAP, Microglia} + Tech | $\hat{\beta}_{\text{Axon}}$ only                                      |
| F8                                                    | GFAP $\rightarrow$ FA (CDE, excl. Cell)                        | GFAP, Cell, Axon, Microglia, Myelin, Tech            | {Cell, Axon, Microglia, Myelin} + Tech            | $\hat{\beta}_{\text{GFAP}}$ only                                      |
| F9                                                    | Microglia $\rightarrow$ FA (CDE, excl. Cell; Iron path active) | Microglia, Cell, Axon, GFAP, Myelin, Tech            | {Cell, Axon, GFAP, Myelin} + Tech                 | $\hat{\beta}_{\text{Microglia}}$ only                                 |
| F10                                                   | Axon $\rightarrow$ FA (interventional mediation)               | G-computation (mediator + outcome models)            | —                                                 | —                                                                     |
| <i>Derived quantities (not separate regressions):</i> |                                                                |                                                      |                                                   |                                                                       |
|                                                       | Axon total indirect effect                                     | —                                                    |                                                   | $\hat{\beta}_{\text{Axon}}^{(F1)} - \hat{\beta}_{\text{Axon}}^{(F7)}$ |
|                                                       | Axon $\rightarrow$ Coherence $\rightarrow$ FA                  | —                                                    |                                                   | $\hat{\beta}_{\text{Axon}}^{(F1)} - \hat{\beta}_{\text{Axon}}^{(F2)}$ |

#### Non-identifiable estimands for FA:

- **NDE/NIE for GFAP  $\rightarrow$  Cell  $\rightarrow$  FA:** Not identified. Latent Oligo violates the no-unmeasured-mediator-outcome-confounder condition for NDE. Report Imai  $\rho$ -sensitivity only.
- **NDE/NIE for Microglia via Cell:** Same issue as GFAP.
- **CDE of Microglia excluding Iron:** Not identified (Iron is latent and unmeasured).
- **Pure total effect of Myelin with only Tech covariates:** Not identified without Cell in the adjustment set (latent Oligo backdoor).
- **Pure total effect of Cell with only its observed parents (Axon, GFAP, Microglia):** Not identified without Myelin in the adjustment set (latent Oligo backdoor).

Table 2: Identifiable estimands for MD models (DAG identical to FA but Coherence node and its edges are absent). Same hierarchical structure, **Tech** = {Fixation Duration, Storage Time}, and identification logic as FA.

| #                        | Estimand                                                       | Predictors in model                       | Adj. set <b>Z</b>                      | Causal coeff.                                                         |
|--------------------------|----------------------------------------------------------------|-------------------------------------------|----------------------------------------|-----------------------------------------------------------------------|
| M1                       | Axon $\rightarrow$ MD (total)                                  | Axon, Tech                                | Tech only                              | $\hat{\beta}_{\text{Axon}}$                                           |
| M2                       | Myelin $\rightarrow$ MD (total)                                | Myelin, Axon, Cell, GFAP, Microglia, Tech | {Axon, Cell, GFAP, Microglia} + Tech   | $\hat{\beta}_{\text{Myelin}}$ only                                    |
| M3                       | GFAP $\rightarrow$ MD (total)                                  | GFAP, Tech                                | Tech only                              | $\hat{\beta}_{\text{GFAP}}$                                           |
| M4                       | Microglia $\rightarrow$ MD (total)                             | Microglia, Tech                           | Tech only                              | $\hat{\beta}_{\text{Microglia}}$                                      |
| M5                       | Cell $\rightarrow$ MD (total)                                  | Cell, Axon, GFAP, Microglia, Myelin, Tech | {Axon, GFAP, Microglia, Myelin} + Tech | $\hat{\beta}_{\text{Cell}}$ only                                      |
| M6                       | Axon $\rightarrow$ MD (CDE, excl. Myelin, Cell)                | Axon, Myelin, Cell, GFAP, Microglia, Tech | {Myelin, Cell, GFAP, Microglia} + Tech | $\hat{\beta}_{\text{Axon}}$ only                                      |
| M7                       | GFAP $\rightarrow$ MD (CDE, excl. Cell)                        | GFAP, Cell, Axon, Microglia, Myelin, Tech | {Cell, Axon, Microglia, Myelin} + Tech | $\hat{\beta}_{\text{GFAP}}$ only                                      |
| M8                       | Microglia $\rightarrow$ MD (CDE, excl. Cell; Iron path active) | Microglia, Cell, Axon, GFAP, Myelin, Tech | {Cell, Axon, GFAP, Myelin} + Tech      | $\hat{\beta}_{\text{Microglia}}$ only                                 |
| M9                       | Axon $\rightarrow$ MD (interventional mediation)               | G-computation (mediator + outcome models) | —                                      | —                                                                     |
| <i>Derived quantity:</i> |                                                                |                                           |                                        |                                                                       |
|                          | Axon total indirect effect                                     | —                                         |                                        | $\hat{\beta}_{\text{Axon}}^{(M1)} - \hat{\beta}_{\text{Axon}}^{(M6)}$ |

**Total model count:** 9 distinct FA regressions + 8 distinct MD regressions + 2 g-computation procedures = 19 fitted models. Derived indirect effects are computed as contrasts between model pairs and from g-computation output.

## 2.6 Critical Identification Warnings

Cell-Density plays two distinct roles depending on the estimand: When Cell is the exposure (F6/M5), its parents (Axon, GFAP, Microglia, Myelin) are confounders and must be controlled for. On the other side, when Myelin or Axon is the exposure, Cell appears in the adjustment set as a proxy-blocker for the latent Oligo backdoor path, not as a mediator (don't interpret this causally in those models).

Myelin-Density also plays two distinct roles: When Myelin is the exposure (F3/M2), then Cell, GFAP, Microglia are confounders needed in **Z**. But when Cell is the exposure (F6/M5), then Myelin appears in **Z** as a proxy for latent Oligo, not as a mediator of Cell.

The latent Oligo creates correlated residuals in the Myelin and Cell models (both have Oligo as an omitted cause). In Bayesian hierarchical models, this manifests as residual posterior correlation between  $\hat{\beta}_{\text{Myelin}}$  and  $\hat{\beta}_{\text{Cell}}$ . (Report these posterior correlations)

## 3 Data Preprocessing

### 3.1 Normalization Strategy

Normalization is applied at the model-input stage (after per-patch mean extraction from the registered histology maps, before PyMC model fitting). The strategy is metric-specific, derived empirically from data characterization of all 13 tissue blocks: intraclass correlation coefficients (ICC), between-block to within-block coefficient-of-variation ratios, and global skewness statistics.

#### Masking and patch inclusion

Each patch must satisfy two criteria to be included in a given model: (1) the *metric-specific artifact mask*, minimum 5 valid voxels) must pass for every metric used in that model (all predictors and the outcome), and (2) the patch must have valid values for all metrics simultaneously (complete-case intersection). The minimum-voxel floor of 5 is a hard quality gate; all retained patches are additionally down-weighted in the likelihood according to their valid-voxel count  $n_i$  (see Section 4.4), so patches with few voxels contribute proportionally less information without biasing point estimates. The metric-specific mask removes only artifacts relevant to each modality individually; a single comprehensive MRI artifact mask would additionally discard patches with valid histology but poor MRI signal, biasing histological predictor means upward (empirically observed 10–14% upward bias in one block). Patches with artifact masks triggered by physically random events (tissue cracks, sectioning bubbles, staining precipitates) are treated as missing at random for the affected metric; any cross-metric missingness (e.g. a crack affecting two metrics derived from the same section) is handled by the complete-case intersection. A sensitivity analysis applies a *full artifact mask* (Section 8).

#### Group 1: FA, MD, Myelin density, Cell density, Axon density, Axon coherence, technical co-variates

No transformation. Global  $z$ -score across all patches, blocks, and ROI types:

$$\tilde{x}_i = \frac{x_i - \bar{x}_{\text{global}}}{\text{SD}(x)_{\text{global}}}. \quad (1)$$

Applies to all metrics that do not require a variance-stabilizing transform: FA, MD, myelin density, cell density, axon density, axon coherence, and the technical covariates (fixation duration, storage time). Between-block variation is real biological signal absorbed by the block random effect; within-block normalization would remove this signal and is explicitly rejected. Global  $z$ -score therefore preserves biologically meaningful between-block variation across all metrics.

#### Group 2: GFAP density, Microglia density (extreme right skew and near-zero mass requiring log-transform)

Variance-stabilizing log-transform, then global  $z$ -score:

$$\tilde{x}_{p,i} = \frac{\log_{10}(x_{p,i} + \varepsilon_p) - \overline{\log_{10}(x_p + \varepsilon_p)}_{\text{global}}}{\text{SD}(\log_{10}(x_p + \varepsilon_p))_{\text{global}}}, \quad (2)$$

where  $\varepsilon_p$  is the first (GFAP) and 5th (Microglia) percentile of all *nonzero* observed values of predictor  $p$  across all blocks (data-adaptive detection floor; computed from strictly positive values to avoid the floor being collapsed to zero by patches with no detected signal). Applies to:

- **GFAP density** (skewness = 1.47, ICC = 0.30): one block with elevated GFAP also shows elevated Microglia, consistent with reactive astrogliosis in an active inflammatory lesion environment—genuine MS biology, not a staining artifact.

- **Microglia density** (skewness = 10.28, ICC = 0.16): extremely sparse, focally clustered distribution ( $\approx 33\%$  of patches report zero detected microglia); the high-density blocks represent genuine active neuroinflammation. Global  $z$ -score (after log-transform) preserves this disease signal.

**Rationale for rejecting within-block alternatives:**

- *Within-block normalization of outcomes* would remove the between-block FA/MD variation that the block random effect is designed to model.
- *Within-block normalization of predictors* is unnecessary: no systematic batch effects are present in the final dataset, and applying within-block normalization would instead remove genuine between-block biological variation.
- *Within-ROI normalization* would remove the between-ROI differences that are the primary signal of interest.

## 4 Statistical Model

### 4.1 General Model Structure

All 17 regression models (9 FA + 8 MD) share a common hierarchical structure; the two g-computation procedures (F10/M9) follow the same likelihood and prior specifications for their component sub-models. We present the general form and then specify the likelihood for FA and MD models separately.

For subregion  $i$  in ROI type  $r$  within block  $b$ , the linear predictor is:

$$\mu_i = \alpha_{r[i]} + \sum_{p=1}^P \beta_{p,r[i]} \tilde{x}_{p,i} + \sum_{k=1}^2 \gamma_k z_{k,i} + v_{b[i]}, \quad (3)$$

where:

- $\alpha_r$  is the ROI-type-specific intercept ( $r = 1, \dots, 10$  for each ROI type),
- $\beta_{p,r}$  is the ROI-type-specific coefficient for the  $p$ -th histological predictor (the number of predictors  $P$  depends on the estimand; see Tables 1 and 2),
- $\tilde{x}_{p,i}$  is the standardized predictor value (globally  $z$ -scored after optional log-transform, depending on the metric; see Section 3.1),
- $\gamma_k$  are fixed-effect coefficients for the two technical covariates: fixation duration and storage time ( $z_{k,i}$ ;  $K = 2$ ),
- $v_b \sim \mathcal{N}(0, \sigma_{\text{block}}^2)$  is the block-level random intercept.

The ROI-type-specific slopes  $\beta_{p,r}$  implement the predictor  $\times$  ROI<sub>type</sub> interaction, yielding tissue-type-specific causal effect estimates without requiring separate models per ROI type.

### 4.2 Likelihood: FA Models

For FA, we adopt a Gaussian likelihood with identity link and heteroscedastic residuals:

$$y_i^{(\text{FA})} \mid \mu_i, \sigma_{r[i]} \sim \mathcal{N}(\mu_i, \sigma_{r[i]}^2), \quad (4)$$

where  $\sigma_r$  is the ROI-type-specific residual standard deviation (see Section 4.4).

**Rationale.** Although FA is theoretically bounded on  $[0, 1]$ , in postmortem tissue the observed values occupy a central range well away from the boundaries, with no ceiling or floor effects. A Gaussian likelihood is simpler, yields directly interpretable coefficients, and is adequate when data do not pile up near bounds. A Beta regression (logit link, mean-precision parameterization) would add complexity without expected benefit. If posterior predictive checks reveal boundary issues, Beta regression can be revisited as a sensitivity analysis.

### 4.3 Likelihood: MD Models

For MD, we adopt a Skew-Normal likelihood applied to the globally  $z$ -scored outcome (see Section 3.1), with a ROI-type-specific skewness parameter:

$$y_i^{(\text{MD})} \mid \mu_i, \sigma_{r[i]}, \alpha_{r[i]} \sim \text{Skew-Normal}(\mu_i, \sigma_{r[i]}, \alpha_{r[i]}), \quad (5)$$

where  $\alpha_r$  is the ROI-type-specific skewness parameter:  $\alpha_r > 0$  encodes a right tail,  $\alpha_r < 0$  a left tail, and  $\alpha_r = 0$  recovers the Normal exactly.

**Rationale.** Data characterization (13 blocks) reveals that the MD distribution differs qualitatively in skew direction across tissue types: white matter sub-patches exhibit a right-skewed distribution, whereas grey matter sub-patches exhibit a left-skewed distribution. A log-Normal likelihood, while appropriate for right-skewed data, fails to accommodate left-skewed grey matter residuals: the log transform monotonically compresses the right side of any distribution and cannot reverse the direction of skew. The Skew-Normal with ROI-type-specific  $\alpha_r$  subsumes the Normal as a special case and naturally handles opposite skew directions within a single, unified likelihood family. Coefficients  $\beta_{p,r}$  retain the same SD-unit interpretation as in the FA models: a one-SD change in predictor  $p$  within ROI type  $r$  corresponds to a  $\beta_{p,r}$ -SD change in standardized MD.

### 4.4 Heteroscedastic Residuals and Patch-Size Weighting

Residual variance is expected to differ across ROI types (e.g., lesion cores are more variable than intact white matter) and to decrease with the number of valid voxels  $n_i$  contributing to each patch mean. We model the effective residual standard deviation for patch  $i$  through an extended log-linear model:

$$\log(\sigma_{\text{eff},i}) = \delta_0 + \delta_{r[i]} + \delta_n \log\left(\frac{n_i}{n_{\text{ref}}}\right), \quad (6)$$

where  $\delta_0$  is a baseline log-SD,  $\delta_r$  represents ROI-type-specific deviations, and  $\delta_n$  captures the dependence on patch size.  $n_{\text{ref}}$  is the median valid-voxel count across all retained patches (a fixed scaling constant). The parameter  $\delta_n$  is expected to be negative: under a pure independent-voxel measurement noise model,  $\delta_n = -0.5$  (i.e.,  $\sigma \propto 1/\sqrt{n}$ ); under pure within-patch biological heterogeneity independent of patch size,  $\delta_n = 0$ . Estimating  $\delta_n$  from data lets the model identify which regime applies. This formulation naturally implements a *soft* patch-size weighting: larger patches are assigned a tighter likelihood, smaller patches a wider one, without any discontinuous threshold behaviour beyond the hard quality floor of 5 valid voxels.

**Prior:**  $\delta_n \sim \mathcal{N}(-0.25, 0.5)$ , weakly informative, slightly negative to reflect the expected direction without forcing the pure-noise model.

Whether ROI-type heteroscedasticity is warranted will be assessed by comparing homoscedastic and heteroscedastic models via LOO-IC and visual residual checks (see Section 7).

### 4.5 Random Effects: Block Hierarchy

Technical confounders (fixation duration and storage time) are constant within each tissue block. While we include these as fixed-effect covariates where values are available, residual block-level variation is captured by the block-level random intercept:

$$v_b \sim \mathcal{N}(0, \sigma_{\text{block}}^2), \quad (7)$$

for  $b = 1, \dots, 13$  blocks. All 13 tissue blocks originate from a single donor; a brain-level random effect is therefore not applicable.

**Design considerations:**

- **Random intercepts only.** Random slopes (e.g., allowing the Myelin  $\rightarrow$  FA effect to vary by block) would probably require substantially more blocks than 13 to estimate reliably and are therefore excluded.
- **No ROI-instance random effect.** Within-ROI variation is the signal, not noise: the ROI  $\times$  predictor interactions capture systematic differences across tissue types.

## 4.6 Handling Collinearity

The DAG-driven approach specifies distinct, principled adjustment sets for each estimand, but several of these sets contain highly correlated variables due to the underlying biological structure. We address this by:

1. Compute variance inflation factors (VIF) for every model in Tables 1 and 2.
2. Report within-ROI pairwise correlations among model predictors alongside each estimand.
3. Where  $VIF > 5$  for the target exposure, report total effects only and flag direct/CDE estimates as unreliable.

## 4.7 Handling Unmeasured Confounders

The DAG analysis (Sections 2 and 2.5) identifies the following roles for unmeasured variables:

- **ECM [latent]:** Extracellular matrix has no measured parents and no children other than FA. It contributes residual FA variance only and does not confound any biological estimand.
- **Hydration-Status [latent]:** Caused by Fixation Duration and Storage Time (measured); affects FA. Because Hydration-Status has no biological predecessors, it does not confound any biological estimand. Fixation Duration and Storage Time are included as technical covariates to proxy for this latent variable.
- **Iron [latent]:** Caused by Microglia-Density; affects FA. It acts as a latent mediator on the Microglia  $\rightarrow$  FA pathway. Its existence means the Microglia CDE excluding Iron cannot be identified without measuring iron content. For total effects of Microglia, Iron does not create a backdoor path (no unmeasured common cause of Microglia and FA beyond Iron itself, which is mediated not confounding).
- **Oligodendrocytes [latent]:** This is the most consequential unmeasured variable. It is a common latent cause of both Myelin-Density and Cell-Density, creating backdoor paths for both exposures. The identification strategy (using observed proxy variables: Cell for the Myelin model; Myelin for the Cell model) is described in Section 2.5. This strategy is valid under linear structural equation assumptions.

Technical factors (Fixation Duration, Storage Time) are constant within tissue blocks and absorbed by block-level random effects plus fixed technical covariates. PMI is identical across all 13 blocks and is excluded from the model.

All assumptions are encoded transparently in the DAG. We will report E-values for the key total-effect estimands to quantify how strong an unmeasured additional confounder (beyond those already modelled) would need to be to explain away the observed effects.

## 4.8 Spatial Dependence

Sub-patches within ROIs are not spatially independent. Adjacent white matter patches, for example, exhibit similar histological composition and similar diffusion properties. Crucially, however, this spatial autocorrelation in the raw data is not a statistical nuisance but a reflection of the biological signal under study. Tissue microstructure varies smoothly across space, and it is precisely this smooth co-variation of histological predictors and diffusion outcomes that our models exploit. To the extent that the spatial correlation is explained by the predictors in the model (myelin density, axon density, etc.), it does not violate model assumptions. The relevant diagnostic is whether residuals remain spatially correlated after conditioning on all predictors.

We deliberately do not include an explicit spatial random effect (e.g., Gaussian process, conditional autoregressive structure) for two reasons. First, a spatial random field captures smooth spatial variation, but so do our histological predictors already. Including both forces them to compete for the same variance, risking attenuation of the very effect estimates we seek to quantify. Second, block-level random effects and ROI-type-specific intercepts already absorb large-scale spatial structure at the block and tissue-compartment level.

We acknowledge that if unmeasured spatially smooth variables (e.g., tissue hydration, ECM) contribute to residual spatial correlation, the effective sample size would be smaller than the nominal count of sub-patches, leading to credible intervals that are somewhat too narrow. This affects precision but not the direction or magnitude of point estimates. We consider this an acceptable trade-off compared to the risk of signal attenuation from an explicit spatial model. To verify this assumption empirically, we will compute residual variograms and Moran’s  $I$  statistics after model fitting. If residuals show negligible spatial structure, the predictors have adequately captured the spatial dependence. If residual spatial correlation is detected, we will report clustered standard errors by connected component as a sensitivity analysis (Section 8), which conservatively inflates credible intervals without introducing a competing spatial field.

## 5 Prior Specification

All priors are weakly informative, chosen to regularize estimates within physically plausible ranges while allowing the data to dominate inference. Table 3 summarizes all prior distributions.

Table 3: Prior distributions for all model parameters.

| Parameter                           | Prior                       | Rationale                                                  |
|-------------------------------------|-----------------------------|------------------------------------------------------------|
| ROI-specific intercepts $\alpha_r$  | $\mathcal{N}(0, 1)$         | Standardized outcome: mean near zero                       |
| Histological effects $\beta_{p,r}$  | $\mathcal{N}(0, 1)$         | $> 2$ SD change per 1 SD predictor is extreme              |
| Technical covariates $\gamma_k$     | $\mathcal{N}(0, 1)$         | Same standardization logic                                 |
| Block RE SD $\sigma_{\text{block}}$ | $\mathcal{N}^+(1)$          | Weakly informative; allows substantial variation           |
| Baseline log-SD $\delta_0$          | $\mathcal{N}(0, 1)$         | Global scale of residual variance                          |
| Heteroscedastic $\delta_r$          | $\mathcal{N}(0, 0.5)$       | Moderate heteroscedasticity on log-scale                   |
| Voxel-count slope $\delta_n$        | $\mathcal{N}(-0.25, 0.5)$   | Slightly negative; expected direction without forcing      |
| MD skewness $\alpha_r$              | $\mathcal{N}(0, 1)$         | Allows right or left skew per ROI type; $\alpha_r = 0$ rec |
| Student- $t$ $\nu$ (sensitivity)    | Gamma(2, 0.1) or fixed at 4 | Prior mass on $\nu = 2$ –30                                |

### 5.1 Justification for Weakly Informative Priors

We do not use literature-informed priors for the following reasons:

1. Prior postmortem MRI–histology studies used different field strengths, resolutions, stains, and tissue types, making effect-size transfer unreliable.
2. An agnostic prior centred at zero allows the data to speak without imposing directional assumptions.
3. Prior sensitivity analysis (Section 8) will verify that results are robust to prior width.

With globally  $z$ -scored predictors and outcomes, the prior  $\mathcal{N}(0, 1)$  implies that a 1 SD change in a predictor producing more than a 2 SD change in the outcome is a priori unlikely but not precluded. This gently regularizes without meaningfully constraining inference.

## 6 Complete Model Specification

We present the full generative model for an FA estimand with  $P$  predictors. MD models share the same linear predictor, random effects, and priors, but use a Skew-Normal likelihood with an

additional ROI-type-specific skewness parameter (see Section 6.2).

### 6.1 FA Model (Gaussian Likelihood)

$$\text{Likelihood: } y_i \mid \mu_i, \sigma_{\text{eff},i} \sim \mathcal{N}(\mu_i, \sigma_{\text{eff},i}^2) \quad (8)$$

$$\text{Linear predictor: } \mu_i = \alpha_{r[i]} + \sum_{p=1}^P \beta_{p,r[i]} \tilde{x}_{p,i} + \sum_{k=1}^2 \gamma_k z_{k,i} + v_{b[i]} \quad (9)$$

$$\text{Effective residual SD: } \log(\sigma_{\text{eff},i}) = \delta_0 + \delta_{r[i]} + \delta_n \log\left(\frac{n_i}{n_{\text{ref}}}\right) \quad (10)$$

$$\text{Fixed-effect priors: } \alpha_r \sim \mathcal{N}(0, 1), \quad \beta_{p,r} \sim \mathcal{N}(0, 1), \quad \gamma_k \sim \mathcal{N}(0, 1) \quad (11)$$

$$\text{Heteroscedastic priors: } \delta_0 \sim \mathcal{N}(0, 1), \quad \delta_r \sim \mathcal{N}(0, 0.5), \quad \delta_n \sim \mathcal{N}(-0.25, 0.5) \quad (12)$$

$$\text{Random effect: } v_b \sim \mathcal{N}(0, \sigma_{\text{block}}^2) \quad (13)$$

$$\text{Hyperprior: } \sigma_{\text{block}} \sim \mathcal{N}^+(1) \quad (14)$$

### 6.2 MD Model (Skew-Normal Likelihood)

The MD model uses the same linear predictor (9), heteroscedastic residual (10), random effect (13), and all priors (11–14) as the FA model, with the likelihood replaced by:

$$\text{Likelihood: } y_i \mid \mu_i, \sigma_{r[i]}, \alpha_{r[i]} \sim \text{Skew-Normal}(\mu_i, \sigma_{r[i]}, \alpha_{r[i]}) \quad (15)$$

$$\text{Skewness prior: } \alpha_r \sim \mathcal{N}(0, 1), \quad r = 1, \dots, R \quad (16)$$

Coefficients  $\beta_{p,r}$  are interpreted on the  $z$ -scored scale: a one-SD change in predictor  $p$  within ROI type  $r$  corresponds to a  $\beta_{p,r}$ -SD change in standardized MD. The posterior distribution of  $\alpha_r$  serves as an empirical check: we expect  $\alpha_r > 0$  for WM ROI types and  $\alpha_r < 0$  for GM ROI types, consistent with the observed skewness pattern.

### 6.3 Sensitivity Model: Student- $t$ Likelihood

For robustness to outliers, a sensitivity analysis replaces the Gaussian likelihood with a Student- $t$  distribution:

$$y_i \mid \mu_i, \sigma_{r[i]}, \nu \sim \text{Student-}t(\nu, \mu_i, \sigma_{r[i]}^2), \quad (17)$$

with either  $\nu$  fixed at 4 (defined variance) or assigned the prior  $\nu \sim \text{Gamma}(2, 0.1)$ . For MD models, the Student- $t$  replaces the Skew-Normal on the  $z$ -scored outcome directly (i.e. the skewness parameter  $\alpha_r$  is dropped and heavier tails are modelled instead). If results are substantively identical to the primary analysis, this confirms robustness to outliers; if they differ, the Student- $t$  version is more trustworthy.

### 6.4 Posterior Computation

All models are fitted using PyMC v5+ with the No-U-Turn Sampler (NUTS). Convergence is assessed via:

- $\hat{R} < 1.01$  for all parameters,
- Effective sample size  $> 400$  per chain,
- Visual inspection of trace plots and rank plots,
- Divergence diagnostics.

Diagnostics are computed using the **ArviZ** library.



## 7 Model Comparison

### 7.1 Information Criteria

Model variants (e.g., homoscedastic vs. heteroscedastic, Gaussian vs. Student- $t$ , with vs. without interaction terms) are compared using leave-one-out information criterion (LOO-IC) computed via Pareto-smoothed importance sampling (PSIS-LOO), implemented in `ArviZ` via `az.loo()`. LOO-IC is preferred over WAIC for hierarchical models due to better behaviour with influential observations.

### 7.2 Posterior Predictive Checks

Visual model adequacy is assessed through:

1. **Density overlays:** Observed outcome distributions versus posterior predictive distributions, overall and stratified by ROI type.
2. **Residual-vs-fitted plots:** To detect systematic patterns indicating model misspecification (e.g., nonlinearity, heteroscedasticity).
3. **Observed-vs-predicted scatter plots:** To assess calibration.

## 8 Sensitivity Analyses

Sensitivity analyses are organised into tiers by priority.

### 8.1 Tier 1: Essential (All Key Estimands)

1. **Likelihood robustness.** Student- $t$  ( $\nu = 4$ ) vs. Gaussian/Skew-Normal for the six highest-priority estimands (F3, F6, F7, M2, M5, M6). Confirms results are not driven by outliers.
2. **Prior sensitivity.**  $\mathcal{N}(0, 1)$  vs.  $\mathcal{N}(0, 2.5)$  vs.  $\mathcal{N}(0, 5)$  on fixed effects for F3, F6, F7, M2, M5, M6. Three prior widths spanning weakly informative to diffuse. Stable posteriors confirm the analysis is data-driven.
3. **Mediation sensitivity.** For all mediation results (F10/M9 and difference-of-coefficient estimates): Imai residual-correlation ( $\rho$ ) sensitivity analysis, reporting the value of  $\rho$  that would reduce the mediated effect to zero. Drop-one-mediator diagnostics for g-computation.
4. **Normalization: within-block  $z$ -score as alternative for Myelin and Cell density.** Refit all models containing Myelin density (F3, F6, F7, F8, F9, M2, M5, M6, M7, M8) and Cell density (F6, F7, F8, F9, M5, M6, M7, M8) using within-block  $z$ -score instead of global  $z$ -score. Stable posteriors confirm that global  $z$ -score adequately preserves biologically meaningful between-block variation.
5. **Normalization: transform choice for Microglia.** Refit models F5/M4 (and F6/M5) using  $\sqrt{x}$  instead of  $\log_{10}(x + \varepsilon)$  for Microglia density. Confirms robustness of results to the variance-stabilizing transform choice.
6. **Patch-size weighting.** Refit F3, F6, F7, M2, M5, M6 with  $\delta_n = 0$  fixed (uniform residual variance, no voxel-count weighting). Confirms that the weighting does not materially shift effect estimates; differences indicate that small patches were disproportionately influencing results.

### 8.2 Tier 2: Important (Selected Representative Models)

9. **Heteroscedastic vs. homoscedastic residuals.** Compare via LOO-IC for representative models F1, F3, F7.
10. **Block as fixed effect vs. random effect.** Verifies that the RE specification adequately captures block-level variation. Run for F3 and M2.

11. **Leave-one-block-out (LOBO) influence analysis.** Refit F3 and M2 dropping one block at a time. Identifies whether any single block drives the results. The high-inflammation block (block\_72\_22\_35) is of particular interest for GFAP and Microglia models.
12. **Proxy validity check for latent Oligo.** A synthetic structural equation model (SEM) with known parameters is used to assess whether the F7 proxy-adjustment strategy — conditioning on Myelin and Cell as proxies for latent Oligodendrocytes — recovers the true Axon  $\rightarrow$  FA controlled direct effect (CDE). The F7 model is fitted to synthetic data generated under a standardised DGP ( $\text{Var}(\text{Myelin}) = \text{Var}(\text{Cell}) = 1$ ,  $\text{Oligo} \rightarrow \{\text{Myelin}, \text{Cell}\}$ ,  $\text{Axon} \rightarrow \text{FA}$  directly). True CDEs are grounded in the F7 posterior means for three target ROI types: WM, WM\_partial\_demyel, and Lesion\_core. A 2-D sensitivity grid is evaluated: (i) *proxy quality* — Oligo–Myelin Pearson correlation  $\gamma_{\text{om}} \in \{0.3, 0.5, 0.7, 0.9\}$  at effect-size scale = 1.0 (4 model fits); and (ii) *effect size* — multiplicative scale  $\in \{0.5\times, 0.75\times, 1.0\times, 1.25\times, 1.5\times\}$  of the real estimates at  $\gamma_{\text{om}} = 0.6$  (5 model fits). Each synthetic dataset contains  $N = 1\,800$  observations (600 per ROI type, balanced). Outcome metrics: absolute bias (posterior mean – true  $\beta_z$ ), relative bias, and 95% HDI coverage of the true CDE per ROI and per grid condition.
13. **Masking: full vs. metric-specific artifact mask.** Refit key models using the full artifact mask instead of the primary metric-specific artifact mask with complete-case intersection.

### 8.3 Tier 3: Deferred

The following are conducted only if requested by reviewers or flagged by diagnostics:

- Spatial structure (clustered standard errors by connected component).
- Outlier handling beyond Student- $t$  (IQR-based removal).
- Microglia  $\varepsilon$  choice:  $\varepsilon = p_1$  (1st percentile of nonzero values) vs. primary  $p_5$  (5th percentile of nonzero values).
- **Multiple imputation of missing histological predictors (MICE).** For patches with a valid outcome but one or more missing predictors due to metric-specific artifacts, impute missing predictor values using Multiple Imputation by Chained Equations ( $M = 10$  datasets). Within-section pairs (GFAP + Cell density; Myelin density + Microglia density) are imputed jointly per section to preserve within-pair correlation. Deferred due to the complexity of joint-missingness patterns from same-section artifact pairs.

### 8.4 Presentation of Sensitivity Results

Key sensitivity results are presented as specification curves or forest plot overlays for selected estimands (e.g., Myelin  $\rightarrow$  FA in WM, Microglia  $\rightarrow$  FA in lesion core), showing effect estimate stability across all Tier 1 and Tier 2 variants.

## 9 Multiple Comparisons

With 17 regression models  $\times R$  ROI types (plus g-computation outputs), the analysis produces a large number of posterior effect estimates (approximately  $17 \times R$  primary coefficients). The following strategy addresses multiplicity:

1. **No formal multiplicity correction.** Bayesian credible intervals are conditional on the model and data (no frequentist family-wise error rate to control). The unified model with ROI  $\times$  predictor interactions applies partial pooling via shared hyperparameters, naturally shrinking extreme estimates. The DAG-driven estimand selection prevents p-hacking, as each model tests a specific, pre-specified causal hypothesis.

2. **Full posterior reporting.** Present complete posterior distributions, credible intervals, and effect sizes. The magnitude and uncertainty of effects speak for themselves rather than being reduced to dichotomous decisions.
3. **Posterior probability of direction (pd).** Reported as a descriptive summary:  $pd > 0.95$  or  $pd > 0.99$  provides a Bayesian analog of directional confidence, used descriptively rather than as a decision threshold.

## 10 Planned Outputs

### 10.1 Main Figures

1. **DAG figure.** DAGs for FA (Figure 1) and MD (Figure 2), with measured and unmeasured variables clearly distinguished.
2. **Effect size heatmap.** ROI types (rows)  $\times$  estimands (columns), colour-coded by posterior mean and annotated for whether the 95% credible interval excludes zero. One panel for FA total effects, one for FA CDEs, one for MD.
3. **Forest plots.** Per estimand, showing ROI-specific coefficients with 95% credible intervals.
4. **Block random effects plot.** Magnitude of block RE standard deviations vs. residual standard deviations vs. fixed effect sizes. Makes the comparison between technical and biological variation explicit.
5. **Posterior predictive check panels.** Observed vs. predicted distributions for representative models, stratified by ROI type.
6. **Specification curve.** For 2–3 key estimands (Myelin  $\rightarrow$  FA, Microglia  $\rightarrow$  FA), showing effect stability across Tier 1 and 2 sensitivity variants.
7. **Mediation decomposition figure.** For Axon  $\rightarrow$  FA: bar/partition plot showing total effect (F1) decomposed into CDE (F7) and indirect contributions per mediator (from g-computation F10), with Imai  $\rho$ -sensitivity overlay.

### 10.2 Main Tables

1. **Sample size table.** ROI type  $\times$  block, showing the number of sub-patches per cell.
2. **Estimand summary table.** All estimands with posterior mean, 95% credible interval, and posterior probability of direction, for each ROI type.
3. **Within-ROI correlation table.** Myelin–Axon correlations by ROI type, justifying where direct effect decomposition is and is not attempted.

### 10.3 Supplementary Material

1. Full posterior distributions for all parameters (ridge plots or violin plots).
2. Leave-one-block-out (LOBO) influence plots.
3. LOO-IC comparison tables across model variants.
4. Residual diagnostic panels (residual-vs-fitted, Q–Q plots) for all regression models.
5. Prior sensitivity results (forest plots across prior widths).
6. G-computation bootstrap distributions and mediator model diagnostics.
7. Imai  $\rho$ -sensitivity curves for GFAP and Microglia mediation.
8. Proxy validity simulation study (synthetic SEM with latent Oligo; F7-model; 2-D grid over Oligo–Myelin proxy quality  $\gamma_{om} \in \{0.3, 0.5, 0.7, 0.9\}$  and effect-size scale  $\in \{0.5 \times - 1.5 \times\}$ ; target ROI types WM, WM\_partial\_demyel, Lesion\_core with true CDEs from F7 posterior means): bias and 95% HDI coverage per ROI and grid condition.
9. **dagitty** code for full DAG reproducibility.
10. Complete PyMC model code.

## 11 Summary of Modelling Choices

Table 4 provides a concise overview of all modelling decisions and their planned sensitivity alternatives.

Table 4: Summary of modelling choices and sensitivity alternatives.

| Choice              | Primary decision                                                                        | Sensitivity alternative                                |
|---------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------|
| FA likelihood       | Gaussian (Normal)                                                                       | Student- $t$ ( $\nu = 4$ )                             |
| MD likelihood       | Skew-Normal<br>(ROI-specific $\alpha_r$ )                                               | Student- $t$ ( $\alpha_r = 0$ fixed,<br>heavier tails) |
| Link function       | Identity (both FA and<br>MD)                                                            | —                                                      |
| Fixed-effect priors | $\mathcal{N}(0, 1)$                                                                     | $\mathcal{N}(0, 2.5)$ , $\mathcal{N}(0, 5)$            |
| RE SD priors        | $\mathcal{N}^+(1)$                                                                      | Exponential(1)                                         |
| Residual variance   | $\log(\sigma) \sim$<br>$\text{ROI}_{\text{type}} + \delta_n \log(n_i/n_{\text{ref}})$   | Homoscedastic; $\delta_n = 0$<br>fixed (Tier 1)        |
| Normalization       | Two groups: global $z$ -score<br>(Group 1); $\log_{10}$ +global<br>$z$ -score (Group 2) | —                                                      |
| Patch inclusion     | <code>metric_artifact_mask</code> ,<br>$n_{\text{min}} = 5$ , complete-case             | <code>full_artifact_mask</code><br>(Tier 2)            |
| Missing predictors  | Complete-case per-model<br>intersection                                                 | MICE, $M = 10$ ,<br>predictor-only (Tier 3)            |
| Block effects       | Random intercept +<br>FixDur, StorTime (PMI<br>excluded: constant)                      | Fixed effect                                           |
| Spatial structure   | Not modelled                                                                            | Clustered SEs (Tier 3)                                 |
| Implementation      | PyMC v5+ (NUTS)                                                                         | —                                                      |